



DAY 36: OBJECT- ORIENTED PROGRAMMING (OOP) IN JAVASCRIPT



Lakshmiprasanna Vakada

What is OOP?

OOP is a programming paradigm based on the concept of "objects", which can hold data (properties) and behaviors (methods).

It helps in writing reusable, scalable, and maintainable code.



Four Pillars of OOP

Let's break down the core principles with simple examples:

1 Encapsulation

Keeping data safe within an object and exposing only what's necessary.

```
class User {  
  #password; // private  
  
  constructor(name, password) {  
    this.name = name;  
    this.#password = password;  
  }  
  
  greet() {  
    console.log(`Hello, ${this.name}`);  
  }  
}
```

2 Abstraction 🧽

Hiding internal details and showing only the essential features.

```
class Car {  
  startEngine() {  
    this.#checkOil();  
    console.log('Engine started!');  
  }  
  
  #checkOil() {  
    console.log('Oil checked!');  
  }  
}
```

3 Inheritance

Allows one class to inherit properties and methods from another.

```
class Animal {  
  speak() {  
    console.log('Animal speaks');  
  }  
}  
  
class Dog extends Animal {  
  bark() {  
    console.log('Dog barks');  
  }  
}  
  
const d = new Dog();  
d.speak(); // Animal speaks  
d.bark();  // Dog barks
```

4 Polymorphism

Different classes can define the same method differently.

```
class Animal {  
  speak() {  
    console.log('Animal speaks');  
  }  
}  
  
class Cat extends Animal {  
  speak() {  
    console.log('Meow');  
  }  
}  
  
const a = new Animal();  
const c = new Cat();  
  
a.speak(); // Animal speaks  
c.speak(); // Meow
```